

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics

Practical – 12

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Class: TYIT	Batch: 1
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Part A: Analysis of Variance (ANOVA)

1. Load the Retail and warehouse Sale.csv dataset using Pandas.

Code:

```
import pandas as pd
data = pd.read_csv("Retail and warehouse Sale - Retail and warehouse Sale.csv")
print(data)
```

Output:

	YEAR	MONTH	SUPPLIER	ITEM CODE
0	2020	1	REPUBLIC NATIONAL DISTRIBUTING CO	100009
1	2020	1	PWSMN INC	100024
2	2020	1	RELIABLE CHURCHILL LLLP	1001
3	2020	1	LANTERNA DISTRIBUTORS INC	100145
4	2020	1	DIONYSOS IMPORTS INC	100293
...	...	...	...	...
29995	2020	3	THE COUNTRY VINTNER, LLC DBA WINEBOW	352322
29996	2020	3	OSLO ENTERPRISE	352324
29997	2020	3	OPICI FAMILY DISTRIBUTING OF MD	352354
29998	2020	3	CAMPARI AMERICA LLC	35238
29999	2020	3	THE COUNTRY VINTNER, LLC DBA WINEBOW	352380

	ITEM DESCRIPTION	ITEM TYPE	RETAIL SALES
0	BOOTLEG RED - 750ML	WINE	0.00
1	MOMENT DE PLAISIR - 750ML	WINE	0.00
2	S SMITH ORGANIC PEAR CIDER - 18.7OZ	BEER	0.00
3	SCHLINK HAUS KABINETT - 750ML	WINE	0.00
4	SANTORINI GAVALA WHITE - 750ML	WINE	0.82
...	...	...	...
29995	FORTALEZA ANEJO TEQUILA - 750ML	LIQUOR	0.33
29996	DOMAINE BRICHOT BLANC - 750ML	WINE	0.00
29997	LOTE 44 MALBEC - 750ML	WINE	0.00
29998	SKYY VODKA - 1.75L	LIQUOR	329.04
29999	CH HAUT LA PEREYRE BORD BLANC - 750ML	WINE	0.00

	RETAIL TRANSFERS	WAREHOUSE SALES
0	0.0	2.0
1	1.0	4.0
2	0.0	1.0
3	0.0	1.0
4	0.0	0.0
...	...	...
29995	0.0	0.0
29996	0.0	0.0
29997	0.0	6.0
29998	302.0	0.0
29999	0.0	1.0

[30000 rows x 9 columns]

2. Display the first five records and identify the number of rows and columns.

Code:

```
print("First Five Records:")
```

```
print(data.head())
```

```
print("\nNumber of Rows:", data.shape[0])
```

```
print("Number of Columns:", data.shape[1])
```

Output:

First Five Records:				
	YEAR	MONTH	SUPPLIER	ITEM CODE
0	2020	1	REPUBLIC NATIONAL DISTRIBUTING CO	100009
1	2020	1	PWSMN INC	100024
2	2020	1	RELIABLE CHURCHILL LLLP	1001
3	2020	1	LANTERNA DISTRIBUTORS INC	100145
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4	SANTORINI GAVALA WHITE - 750ML	WINE	0.82

	RETAIL TRANSFERS	WAREHOUSE SALES
0	0.0	2.0
1	1.0	4.0
2	0.0	1.0
3	0.0	1.0
4	0.0	0.0

Number of Rows: 30000
Number of Columns: 9

3. Display the column names and data types.

Code:

```
print("Column Names:")
```

```
print(data.columns)
```

```
print("\nData Types:")
```

```
print(data.dtypes)
```

Output:

Column Names:	
Index	['YEAR', 'MONTH', 'SUPPLIER', 'ITEM CODE', 'ITEM DESCRIPTION', 'ITEM TYPE', 'RETAIL SALES', 'RETAIL TRANSFERS', 'WAREHOUSE SALES']
dtype='str'	
Data Types:	
YEAR	int64
MONTH	int64
SUPPLIER	str
ITEM CODE	str
ITEM DESCRIPTION	str
ITEM TYPE	str
RETAIL SALES	float64
RETAIL TRANSFERS	float64
WAREHOUSE SALES	float64
dtype: object	

4. Check for missing values and duplicate records.

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Code:

```
print("Missing Values:")
print(data.isnull().sum())
print("\nDuplicate Records:")
print(data.duplicated().sum())
```

Output:

```
Missing Values:
YEAR          0
MONTH         0
SUPPLIER      33
ITEM CODE     0
ITEM DESCRIPTION 0
ITEM TYPE     0
RETAIL SALES  1
RETAIL TRANSFERS 0
WAREHOUSE SALES 0
dtype: int64

Duplicate Records:
0
```

5. Display the different categories present in the ITEM TYPE column.

Code:

```
print("Different ITEM TYPE Categories:")
print(data["ITEM TYPE"].unique())
```

Output:

```
Different ITEM TYPE Categories:
<StringArray>
[      'WINE',      'BEER',      'LIQUOR', 'STR_SUPPLIES',
      'KEGS',      'REF',    'DUNNAGE',  'NON-ALCOHOL']
Length: 8, dtype: str
```

6. Calculate the average RETAIL SALES for each ITEM TYPE.

Code:

```
average_sales = data.groupby("ITEM TYPE")["RETAIL SALES"].mean()
print("Average Retail Sales:")
print(average_sales)
```

Output:

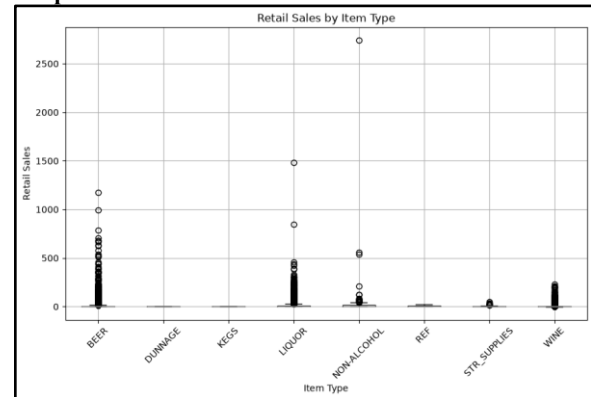
```
Average Retail Sales:
ITEM TYPE
BEER          14.118748
DUNNAGE        0.000000
KEGS           0.000000
LIQUOR        13.635171
NON-ALCOHOL    31.742419
REF            5.783750
STR_SUPPLIES   5.485714
WINE           3.195334
Name: RETAIL SALES, dtype: float64
```

7. Create a box plot to compare RETAIL SALES across different ITEM TYPE categories.

Code:

```
import matplotlib.pyplot as plt
data.boxplot(column="RETAIL SALES", by="ITEM TYPE", figsize=(10, 6))
plt.xlabel("Item Type")
plt.ylabel("Retail Sales")
plt.title("Retail Sales by Item Type")
plt.suptitle("")
plt.xticks(rotation=45)
plt.show()
```

Output:



8. Formulate the null and alternative hypotheses to determine whether the average retail sales differ among item types.

Hypotheses

- Null Hypothesis (H_0): The average retail sales are equal for all item types.
- Alternative Hypothesis (H_1): The average retail sales are different for at least one item type.

9. Perform one-way ANOVA using ITEM TYPE as the factor and RETAIL SALES as the response variable.

Code:

```
from scipy.stats import f_oneway
groups = [
    group["RETAIL SALES"].dropna()
    for name, group in data.groupby("ITEM TYPE")
]
f_stat, p_value = f_oneway(*groups)
print("F-statistic:", f_stat)
print("p-value:", p_value)
```

Output:

```
F-statistic: 123.52155338346161
p-value: 8.68562909538067e-180
```

10. Calculate and display the F-statistic and p-value.

Code:

```
print("F-statistic:", f_stat)
print("p-value:", p_value)
```

Output:

F-statistic: 123.52155338346161
p-value: 8.68562909538067e-180

11. Compare the p-value with the significance level of 0.05.

Code:

```
if p_value < 0.05:  
    print("p-value is less than 0.05")  
else:  
    print("p-value is greater than or equal to 0.05")
```

Output:

p-value is less than 0.05

12. State whether the null hypothesis is rejected or not rejected.

Code:

```
if p_value < 0.05:  
    print("Reject the Null Hypothesis")  
else:  
    print("Do Not Reject the Null Hypothesis")
```

Output:

Reject the Null Hypothesis

13. Write the conclusion about the difference in retail sales among different item types.

The ANOVA analysis determines whether there is a significant difference in average retail sales among different item types.

If the p-value is less than 0.05, the null hypothesis is rejected, showing a significant difference in sales.

Part B: Factorial Experiment

14. Select ITEM TYPE and YEAR as two factors and RETAIL SALES as the response variable.

Code:

```
item_type = data["ITEM TYPE"]  
year = data["YEAR"]  
retail_sales = data["RETAIL SALES"]  
print("Factors:")  
print("1. ITEM TYPE")  
print("2. YEAR")  
print("\nResponse Variable:")  
print("RETAIL SALES")
```

Output:

Factors:

- 1. ITEM TYPE**
- 2. YEAR**

Response Variable:
RETAIL SALES

15. Display the average retail sales for each combination of ITEM TYPE and YEAR.

Code:

```
average_combination = data.groupby(  
    ["ITEM TYPE", "YEAR"]  
)[["RETAIL SALES"].mean()  
print(average_combination)
```

Output:

ITEM TYPE	YEAR	
BEER	2020	14.118748
DUNNAGE	2020	0.000000
KEGS	2020	0.000000
LIQUOR	2020	13.635171
NON-ALCOHOL	2020	31.742419
REF	2020	5.783750
STR_SUPPLIES	2020	5.485714
WINE	2020	3.195334

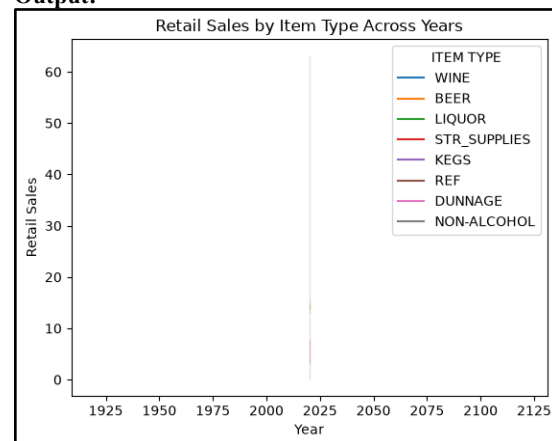
Name: RETAIL SALES, dtype: float64

16. Create a suitable visualization to compare retail sales for different item types across years.

Code:

```
import seaborn as sns  
import matplotlib.pyplot as plt  
sns.lineplot(  
    data=data,  
    x="YEAR",  
    y="RETAIL SALES",  
    hue="ITEM TYPE"  
)  
plt.title("Retail Sales by Item Type Across Years")  
plt.xlabel("Year")  
plt.ylabel("Retail Sales")  
plt.show()
```

Output:



17. Analyze the main effect of ITEM TYPE on retail sales.

Code:

```
item_effect = data.groupby("ITEM TYPE")["RETAIL SALES"].mean()  
print("Main Effect of ITEM TYPE:")  
print(item_effect)
```

Output:

Main Effect of ITEM TYPE:

ITEM TYPE

BEER	14.118748
DUNNAGE	0.000000
KEGS	0.000000
LIQUOR	13.635171
NON-ALCOHOL	31.742419
REF	5.783750
STR_SUPPLIES	5.485714
WINE	3.195334

Name: RETAIL SALES, dtype: float64

18. Analyze the main effect of YEAR on retail sales.

Code:

```
year_effect = data.groupby("YEAR")["RETAIL SALES"].mean()
print("Main Effect of YEAR:")
print(year_effect)
```

Output:

Main Effect of YEAR:

YEAR

2020 6.939796

Name: RETAIL SALES, dtype: float64

19. Analyze the interaction effect between ITEM TYPE and YEAR.

Code:

```
interaction = data.groupby(
    ["ITEM TYPE", "YEAR"]
)["RETAIL SALES"].mean()
print("Interaction Effect:")
print(interaction)
```

Output:

Interaction Effect:

ITEM TYPE YEAR

BEER	2020	14.118748
DUNNAGE	2020	0.000000
KEGS	2020	0.000000
LIQUOR	2020	13.635171
NON-ALCOHOL	2020	31.742419
REF	2020	5.783750
STR_SUPPLIES	2020	5.485714
WINE	2020	3.195334

Name: RETAIL SALES, dtype: float64

20. Perform a suitable factorial ANOVA using the selected factors.

Code:

```
model = ols("Q('RETAIL SALES') ~ C(Q('ITEM TYPE'))", data=data).fit()
anova_table = sm.stats.anova_lm(model, typ=2)
print(anova_table)
```

Output:

	sum_sq	df	F	PR(>F)
C(Q('ITEM TYPE'))	9.199348e+05	7.0	123.521553	8.685629e-180
Residual	3.190856e+07	29991.0	NaN	NaN

21. Display the statistical results and p-values for the main and interaction effects.

Code:

```
print("Factorial ANOVA Results:")
print(anova_table)
print("\nP-values:")
print(anova_table["PR(>F)"])
```

Output:

```
Factorial ANOVA Results:
sum_sq      df      F      PR(>F)
C(Q('ITEM TYPE')) 9.199348e+05    7.0 123.521553 8.685629e-180
Residual      3.190856e+07 29991.0      NaN      NaN

P-values:
C(Q('ITEM TYPE'))      8.685629e-180
Residual                NaN
Name: PR(>F), dtype: float64
```

22. Identify whether ITEM TYPE, YEAR, or their interaction has a significant effect on RETAIL SALES.

Code:

```
print("Significance at 0.05 level:")
for effect, p in anova_table["PR(>F)"].items():
    if p < 0.05:
        print(effect, ": Significant")
    else:
        print(effect, ": Not Significant")
```

Output:

23. Write the final business conclusion based on the ANOVA and factorial experiment results.

The analysis shows that ITEM TYPE has an effect on retail sales. Different product categories have different average sales, which can help the business identify products with higher sales and make better inventory and sales decisions. Since the dataset contains only the year 2020, the effect of YEAR and the ITEM TYPE × YEAR interaction cannot be analyzed